

changes with minimal disruption to service. Among the many applications of Docker, one of the most common is the transition from monolithic to microservices based architectures.

One of the most interesting applications of microservices architecture is the container. A container is a standardised software package that separates an application's runtime from the system on which it runs by enclosing it with all of its dependencies and resources. Containers are different from virtual machines in that they only virtualise software layers above the OS.

Docker's positive impact on enterprises and real world scenarios

Because of its mobility and ecosystem, Docker has grown to be synonymous with containerization. It has been adopted and supported by all the main cloud providers including GCP, AWS, Azure and many others. Organisations are saving a lot of money by using containerization and running several containers on a single host. Since Docker enables service deployment on inexpensive commodity hardware, it may be used to avoid spending money on costly servers.

Docker is prominently used for creating and managing containerized development environments with Docker images, which enable rapid provisioning and consistent performance over the course of a project's life span. In addition, it provides comprehensive assistance for all development-related tools, frameworks, and technologies.

Docker environments also make it easier to use tools like CI/CD, A/B testing, and webhooks. Thus, it is simple to include Bitbucket or GitHub repos with the development environment, allowing for automated builds to be generated from the source code and afterwards pushed to the Docker Repo.

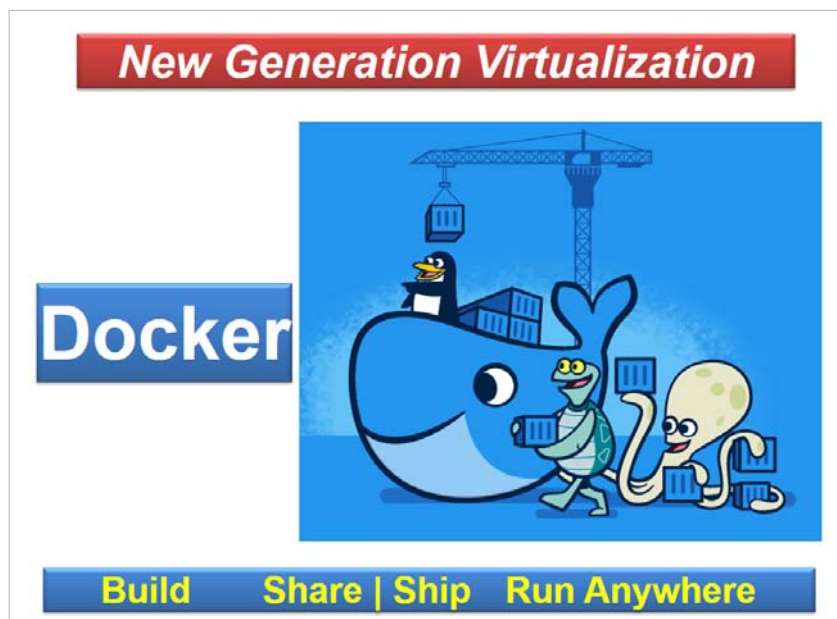


Figure 1: Docker platform

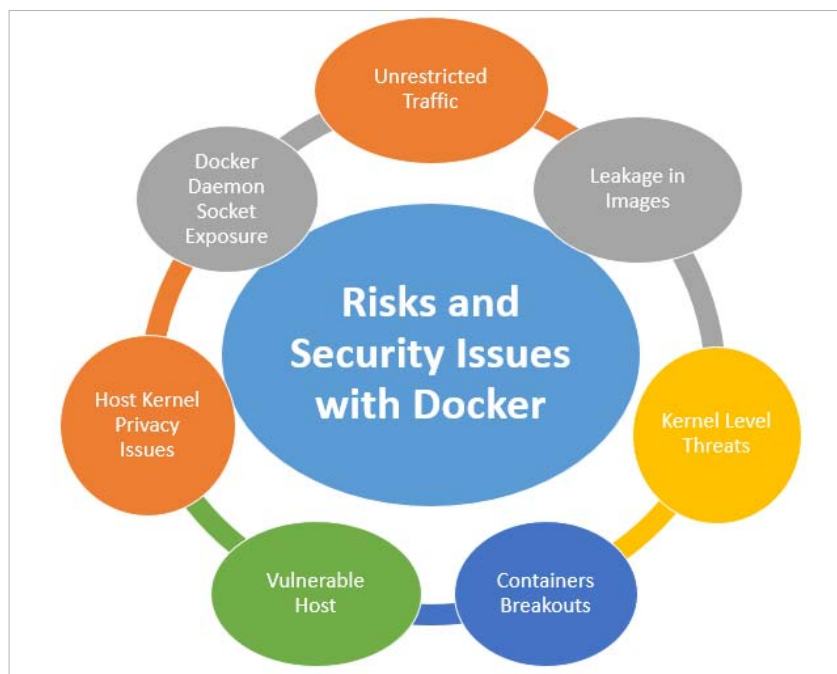


Figure 2: Key risks and security issues with Docker

One of the most important ways to put Docker to use is in the development and deployment of software. There is less risk of unexpected issues arising due to configuration variations between the test environment and the production environment when developers conduct

tests in the same environment where the apps will run in production.

Security and integrity aspects in Docker containers

Once attackers have compromised a host, they frequently have the resources to go laterally and compromise more